Modern Academy
Engineering and Technology in Maadi
Electronic Engineering and
Communications Technology Department

Academic Year 2022-2023
Semester Summer
Exam Date 31 / 7/ 2023

QUESTIONS FOR THE MID TERM EXAM

Subject: (ELCN425 – ELC 422) Digital Signal Processing
Examiners: Dr. Samir M. Kamal & Dr. Sara M. Hassan
Spec.: Level- 4, Elec. Eng. & Comm. Tech. Dep.
Time: 50 Min

Number of Pages: one Model 2 Attempt all questions

Question 1 [20 Points]

a- In detail, what are the main characteristics of the discrete time sinusoidal signals?
b- Check the periodicity of the following signal, $x_a(t) = 3\cos(5t + \pi/6)$
c- For the signal $x(t) = \cos(200\pi t)$ to be sampled with sampling frequency $F_S = 400$Hz.
(i) Find an expression for its corresponding sequence, x(n). Then calculate its fundamental period N_P.
(ii) Find the samples values of the first cycle in x(n), obtained in part (i), then plot x(n) and $x(t)$ at the same figure.

Question 2 [20 Points]

a- In sampling process, drive the sampling condition.
b- For the following sequence: $x(n) = \{1, 0, 1, 1\}$:
(i) Using FFT algorithm find the frequency components of x(n), X(k),
(ii) Check your results, obtained in part (i), by calculating x(n) using IDFT method.
(iii) Using your results, verify Parseval's theorem.
(iv) Plot the amplitude and the phase spectrums.
(v) What is the computational saving when using FFT algorithm instead of DFT algorithm?

GOOD LUCK